

Implicit Differentiation and Inverse Functions



Implicit differentiation basics

- 1 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.
 - 2 Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
 - 3 Find $\frac{dy}{dx}$ for $\sin(xy) = x$.
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Tangent lines to implicit curves

- 4 Find the equation of the tangent line to $x^2 + y^2 = 25$ at the point $(3, 4)$.
 - 5 Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $x^2y + y^3 = 10$.
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Second derivatives via implicit differentiation

- 6 Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 1$, expressing your answer in terms of x and y .
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Derivatives of inverse functions

- 7 If $f(x) = x^3 + 2x + 1$, find $(f^{-1})'(4)$, using $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ and noting $f(1) = 4$.
 - 8 State the derivatives of the inverse trigonometric functions $\sin^{-1}x$, $\cos^{-1}x$, and $\tan^{-1}x$.
 - 9 Differentiate $f(x) = \tan^{-1}(3x)$.
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Implicit differentiation with logarithmic and exponential terms

- 10 Find $\frac{dy}{dx}$ for $e^y = xy$.
 - 11 Find $\frac{dy}{dx}$ for $\ln(x + y) = x^2$.
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Related applications of implicit differentiation

12 Find all points on the curve $x^2 + xy + y^2 = 7$ where the tangent line is horizontal.